

Question 1: An RGB color image has a spatial resolution of 640×480 pixels.

Each pixel is represented using 8 bits per color channel (R, G, and B).

1. Convert the given RGB pixel ($R = 120$, $G = 200$, $B = 80$) into its equivalent grayscale value using the standard luminance equation. (3 marks)

2. Convert the same RGB pixel ($R = 120$, $G = 200$, $B = 80$) into the HSV color model.

Clearly show the calculation of Hue, Saturation, and Value. (4 marks)

3. Calculate the total size of the RGB image in megabytes (MB) before compression. (2 marks)

4. After applying an image compression technique, the image size is reduced to 450 KB.

Calculate the compression ratio. (2 marks)

5. If the image is transmitted at 25 frames per second,

determine the required bit rate for transmission of the compressed image. (1 mark)

Question 2: A grayscale digital image has a spatial resolution of 1024×768 pixels and is represented using 6 bits per pixel.

1. Calculate the total number of pixels in the image. (2 marks)

2. Determine the gray level resolution of the image and specify the intensity range. (2 marks)

3. Calculate the total memory required to store the image in kilobytes (KB) and megabytes (MB). (3 marks)

4. The image is obtained by sampling a continuous-time image signal having a maximum frequency of 4 kHz.

• Check whether a sampling frequency of 7 kHz will cause aliasing. Justify your answer using the sampling theorem. (3 marks)

5. Explain 4-connected, 8-connected, and m-pixel connectivity diagrams. (2 marks)

connected, 8- and m-pixel with neat marks)

Question 3:

Consider the following two discrete-time signals:

$$x(n) = \{1, 2, 1\}$$

$$h(n) = \{2, 1, -1\}$$

1. Compute the linear convolution of $x(n)$ and $h(n)$. Show all steps. (3 marks)

2. Find the 3-point circular convolution of $x(n)$ and $h(n)$. (2 marks)

3. Calculate the auto-correlation of the signal $x(n)$. (2 marks)

4. A 3×3 grayscale image segment is given below:

$$\begin{bmatrix} 100 & 120 & 140 \\ 110 & 130 & 150 \\ 120 & 140 & 160 \end{bmatrix}$$

Apply a 3×3 mean filter and determine the output value of the center pixel. (2 marks)

5. The histogram of a grayscale image has 64 gray levels and 4096 total pixels uniformly distributed.

• Perform histogram equalization and find the new gray level corresponding to original gray level 20. (2 marks)

6. State one difference between linear convolution and circular convolution. (1 mark)

Question 4: Design a digital low-pass FIR filter using the window method to meet the following specifications:

- Passband edge frequency, $f_p = 1$ kHz
 - Stopband edge frequency, $f_s = 1.5$ kHz
 - Sampling frequency, $f_{\text{sampling}} = 8$ kHz
 - Minimum stopband attenuation = **40 dB**
1. **Determine the normalized digital passband and stopband frequencies.** (2 marks)
 2. **Select a suitable window function** to satisfy the given attenuation requirement and justify your choice. (2 marks)
 3. **Calculate the transition width and estimate the order of the FIR filter.** (3 marks)
 4. **Write the expression for the ideal impulse response of the low-pass filter.** (2 marks)
 5. **Explain how the selected window modifies the ideal impulse response to obtain the practical FIR filter.** (2 marks)
 6. **State one advantage of FIR filters over IIR filters.** (1 mark)

Question 5:

A binary image A and a structuring element B are given below:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

1. **Perform the dilation operation $A \oplus B$ and obtain the output image.** (3 marks)
2. **Perform the erosion operation $A \ominus B$ and obtain the output image.** (3 marks)
3. **Find the opening of the image $A \circ B$.** (2 marks)
4. **Find the closing of the image $A \bullet B$.** (2 marks)
5. **Explain boundary extraction and write the expression used for boundary extraction in morphological processing.** (1 mark)
6. **State one application of morphological image processing in segmentation.** (1 mark)

Question 6:

Consider the discrete-time signal

$$x(n) = \{1, 0, 2, 0\}, n = 0, 1, 2, 3$$

1. **Compute the 4-point Discrete Fourier Transform (DFT) of the signal $x(n)$.**

Show all steps. (4 marks)

2. **Find the 4-point Inverse Discrete Fourier Transform (IDFT)** of the result obtained in part (1) and verify that the original signal is recovered. (2 marks)
3. **Perform the 4-point Fast Fourier Transform (FFT)** of $x(n)$ using the **Decimation-in-Time (DIT)** algorithm.

Draw the butterfly diagram and compute the final FFT output. (3 marks)

4. **Verify the linearity property of the DFT** using the signals

$$x_1(n) = \{1, 0, 1, 0\} \text{ and}$$

$$x_2(n) = \{0, 1, 0, 1\}. \text{ (2 marks)}$$

5. **State one advantage of FFT over direct DFT computation.** (1 mark)

Color image Processing & Compression

Q.1) image resolution = 640×480 pixels

bits per color channel = 8 bits

RGB pixel = (R=120, G=200, B=80)

Compressed image size = 450 KB

Frame rate = 25 frames/sec

Find

a) RGB $\rightarrow$ Gray conversion

Standard luminance equation

$$\text{Grayscale} = 0.299R + 0.587G + 0.114B$$

$$= 0.299(120) + 0.587(200) + 0.114(80)$$

$$= 35.88 + 117.4 + 9.12$$

$$231.4 \approx 162.5 \approx 162 \times 0.001 = 162 \times 10^{-3}$$

b) RGB $\rightarrow$ HSV conversion

Normalize RGB values

$$R = \frac{120}{255} = 0.47, \quad G = \frac{200}{255} = 0.78, \quad B = \frac{80}{255} = 0.31$$

$$\text{Max} = 0.78, \quad \text{Min} = 0.31$$

$$\Delta = \text{max} - \text{min} = 0.47$$

$$\text{saturation} = \frac{\Delta}{\text{max}} = \frac{0.47}{0.78} = 0.60$$

Hue (H):

max = G

$$H = 60^\circ \times \left[2 + \frac{(B-R)}{0.47} \right]$$

$$= 60 \times \left[2 + \frac{0.31 - 0.47}{0.47} \right]$$

$$= 60 \times (2 - 0.34)$$

$$= 99.6^\circ$$

c) image size before compression

$$\text{total pixels} = 640 \times 480 = 307,200 \text{ samples}$$

$$\text{Bit Per Pixel} = 8 \times 3 = 24 \text{ bits}$$

$$\text{Total bits} = 307,200 \times 24 = 7,372,800 \text{ bits}$$

$$\text{convert into MB: Bytes} = \frac{7,372,800}{8} = 921,600$$

$$\text{MB} = \frac{921,600}{1024^2} = 0.88 \text{ MB}$$

d) compression Ratio

$$= \frac{\text{original size}}{\text{compressed size}}$$

convert original size to KB $= 0.88 \times 1024$
 $= 901 \text{ KB}$

$$CR = \frac{901}{450} \approx 2.0022$$

e) Bit Rate Calculation

convert compressed size to bits 450 KB

$$450 \times 1024 \times 8 = 3686400 \text{ bits}$$

$$\text{Bit rate} = 3686400 \times 25$$

$$= 92160000 \text{ bit/sec}$$

$$= 892.16 \text{ mbps}$$

Q.2) Spatial Resolution = 1024×768 Pixels
 Bits Per Pixels = 6 bits
 max signal Fz. = 4 kHz
 sampling Fz. = 7 kHz

c) total number of pixels = $1024 \times 768 = 786432$ pixels

b) Gray level = $2^6 = 2^6 = 64$

Intensity range: 0 to (64-1)

: 0 to 63

64 levels

c) memory required to store image

$$\text{Total bits} = 786432 \times 6$$

$$= 4718592 \text{ bits}$$

convert bit to byte

$$= \frac{4718592}{8}$$

$$= 589824 \text{ bytes}$$

$$\text{Convert KB} = \frac{589824}{1024} = 576 \text{ KB}$$

$$\text{Convert MB} = \frac{576}{1024} = 0.5625 \text{ MB}$$

d) aliasing check using sampling theorem.

$$f_s \geq 2f_{\max}$$

$$2f_{\max} = 2 \times 4 = 8 \text{ KHz}$$

$$7 \text{ KHz} \leq 8 \text{ KHz}$$

e) Pixel connectivity = 4 = 4 = 4 = 4

- 4 connected (Pixel)

$$p - \begin{matrix} p \\ | \\ p \end{matrix} - p$$

- 8 connected Pixels

$$\begin{matrix} p & p & p \\ p & x & p \\ p & p & p \end{matrix}$$

$$(x-m) \times (n-l) \times \dots = \dots$$

$$d = s_0 + s_1 + s_2 = 102$$

$$u = (1)(5) + (5)(1) = 102$$

$$l = (1)(1) = 102$$

Q.3) $x(n) = \{1, 2, 1\}$
 $h(n) = \{2, 1, -1\}$

$$x(n), h(n), d(n), u(n), l(n) = \dots$$

a) Linear convolution

length $L = 3 + 3 - 1 = 5$

$$\begin{matrix} \begin{matrix} 001 & 091 & 001 \\ 021 & 081 & 011 \\ 1001 & 001 & 051 \end{matrix} \end{matrix} = \dots$$

$$y(0) = 1 \cdot 2 = 2$$

$$y(1) = 1 \cdot 1 + 2 \cdot 2 = 5$$

$$y(2) = 1 \cdot (-1) + 2 \cdot 1 + 1 \cdot 2$$

$$= -1 + 2 + 2 = 3$$

$$y(3) = 2 \cdot (-1) + 1 \cdot 1 = -2 + 1 = -1$$

$$y(4) = 1 \cdot (-1) = -1$$

$$\frac{0511}{p}$$

$$y(n) = \{2, 5, 3, -1, -1\}$$

b) 3 point circular convolution

$$y(n) = \sum_{k=0}^{2} x(k) h(n-k) \text{ mod } 3$$

$$\dots$$

$$y(0) = 1(2) + 2(-1) + 1(1) = 1$$

$$y(1) = 1(1) + 2(2) + 1(-1) = 4$$

$$y(2) = 1(-1) + 2(1) + 1(2) = 3$$

$$y(n) = \{1, 4, 3\}$$

c) auto correlation of $x(n)$

$$R_{xx} = \sum x(n) x(n-k)$$

$$R(0) = 1^2 + 2^2 + 3^2 = 14$$

$$R(1) = (1)(2) + (2)(1) = 4$$

$$R(2) = (1)(1) = 1$$

$$R_{xx} = \{14, 4, 1\}$$

d) mean filter operation

$$\text{image} = \begin{bmatrix} 100 & 120 & 140 \\ 110 & 130 & 150 \\ 120 & 140 & 160 \end{bmatrix}$$

$$\text{mean filter} = \frac{\text{sum of pixel}}{3 \times 3}$$

$$= \frac{1170}{9}$$

$$= 130 = \text{center pixel value}$$

e) Histogram Equalization

$$\text{Gray level} = 64, (x-m) \text{ and } (x) \text{ etc. } \sum_{i=0}^{63} 1 = 64$$

$$\text{total pixels} = 4096$$

$$\text{pixels per gray level} = \frac{4096}{64} = 64$$

$$P = 1 - (1) + 0.4 + (1) = 1.4$$

$$E = (1.64) + (1) = 2.64$$

$$E = 1.64 + 1 = 2.64$$

$$CDF = (20 + 1) \times 64 = 1344$$

CDF up to gray level 20: 1344

$$CDF = (20 + 1) \times 64 = 1344$$

$$S_k = \frac{CDF}{\text{Total Pixel}} \times (L-1)$$

$$= \frac{1344}{4096} \times 63 \approx 20$$

new gray level: 20

Q.4) Passband edge frequency: $f_p = 1 \text{ kHz}$

Stopband edge frequency: $f_s = 1.5 \text{ kHz}$

sampling $f_s = f_{sa} = 8 \text{ kHz}$

min stopband attenuation = 40 dB

a) normalized digital passband and stopband f_{no}

$$\omega = \frac{2\pi f}{f_{sam}}$$

$$\text{Passband } f_{pe} = \omega_p = \frac{2\pi \times 1000}{8000} = \frac{\pi}{4} \text{ rad/sample}$$

$$\text{Stopband } f_{se} = \omega_s = \frac{2\pi \times 1500}{8000} = \frac{3\pi}{8} \text{ rad/sample}$$

$$\left. \begin{aligned} \sin \alpha &= \frac{(s16 - \alpha) \pi}{(s16 - \alpha) \pi} \\ \sin \alpha &= \alpha \end{aligned} \right\} = (1) \text{ rad}$$

b) selection of window function

- required stopband attenuation 40 dB

Hamming window 41 dB attenuation

c) Transition width and filter order

Transition width: $\Delta\omega = \omega_s - \omega_p$

$$= \frac{3\pi}{8} \frac{\pi}{4}$$

$$\Delta\omega = \pi/8 \text{ rad/s}$$

- FIR filter order (Hamming window)

$$N \approx \frac{3.3\pi}{\Delta\omega} = \frac{3.3\pi}{\pi/8} \approx 26.4$$

Filter length $= N+1 = 28$

d) Ideal impulse response of low pass filter

cut off freq.

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{\pi/4 + 3\pi/8}{2} = \frac{5\pi}{16}$$

ideal impulse response

$$h_d(n) = \begin{cases} \frac{\sin(n\omega_c/2)}{n\omega_c/2}, & n \neq N/2 \\ \omega_c/2, & n = N/2 \end{cases}$$

e) effect of window on ideal impulse response.

- infinite length

f) Advantages of FIR over IIR Filter.

- exact linear phase

Ques) Binary image $A = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Structure using element $B(3 \times 3)$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

a) Division $(A \oplus B)$

Rule: sets the output pixels to 1 if any pixel under B is 1.

$$A \oplus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

b) Erosion $A \ominus B$

Rule: Erosion sets the output pixel to 1 only if all pixels under B are 1.

$$A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

c) Opening $A \circ B$

$$A \circ B = (A \ominus B) \oplus B \quad \because A \ominus B = 0$$

$$A \circ B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \oplus B = B$$

d) Closing $A \bullet B$

$$A \bullet B = (A \oplus B) \ominus B$$

$$A \bullet B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \ominus B = B \oplus A$$

e) Boundary Extraction

- finds the outlines of objects in a binary image.

$$\text{Boundary}(A) = A - (A \ominus B)$$

Q.6]

$$x(n) = \{1, 0, 2, 0\} \quad n = 0, 1, 2, 3$$

Number of points $(N = 4)$

$$\text{Let } \omega_4 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = -j$$

a) 4 Point DFT of $x(n)$

DFT Formula: $X(k) = \sum_{n=0}^{N-1} x(n) \omega_N^{kn}$

$$X(k) = \sum_{n=0}^3 x(n) \omega_4^{kn}$$

$$k=0 \quad x(0) = 1+0+2+0 = 3$$

$$k=1 \quad x(1) = 1 + 2\omega_4^2 = e^{-j\pi} = -1$$

$$= 1 + 2(-1) = -1$$

$$k=2 \quad x(2) = 1 + 2\omega_4^4 = 1 + 2 = 3$$

$$k=3 \quad x(3) = 1 + 2\omega_4^6$$

$$= 1 - 2 = -1$$

$$X(k) = \{3, -1, 3, -1\}$$

b) 4 point IDFT and verification

IDFT Formula

$$x(n) = \frac{1}{4} \sum_{k=0}^3 x(k) W_4^{-kn}$$

$$x(k) = \{3, -1, 3, -1\}$$

$$x(0) = \frac{1}{4} (3 + 1 + 3 + 1) = 2$$

$$x(1) = \frac{1}{4} (3 + j - 3 - j) = 0$$

$$x(2) = \frac{1}{4} (3 + 1 + 3 + 1) = 2$$

$$x(3) = \frac{1}{4} (3 - j - 3 + j) = 0$$

$$x(n) = \{2, 0, 2, 0\}$$

c) 4 point FFT using DIT

$$E = 0 + 5 + 10 + 15 = (0)X \quad 0 = 1$$

- separate even and odd samples

$$F = 1 + 4 + 9 + 16 = (1)X \quad 1 = 2$$

$$x_e = \{1, 2\} \quad x_o = \{0, 3\}$$

- 2 point DFTs

$$G = 0 + 3 + 6 + 9 = (3)X \quad 3 = 4$$

$$H = 1 + 2 =$$

$$x_e = \{1 + 2, 1 - 2\} = \{3, -1\}$$

$$I = 0 + 1 + 4 + 9 = (8)X \quad 8 = 4$$

$$x_o = \{0, 3\}$$

$$J = 1 - 2 =$$

$$K = 1 - 4 + 9 - 16 = (-1)X$$

- Combine

$$x(k) = x_e(k) + w_k x_o(k)$$

$$x(0) = 3, x(1) = -1$$

$$x(2) = 3$$

$$x(3) = -1$$

$$\text{FFT output} = \{3, -1, 3, -1\}$$

d) verification of DFT linearity

$$x_1(n) = \{1, 0, 1, 0\}$$

$$x_2(n) = \{0, 1, 0, 1\}$$

DFTs:

$$\text{DFT}[x_1] = \{2, 0, 2, 0\}$$

$$\text{DFT}[x_2] = \{2, 0, -2, 0\}$$

$$\text{DFT}[x_1] + \text{DFT}[x_2] = \{4, 0, 0, 0\}$$

$$\text{sum signal: } x_1 + x_2 = \{1, 1, 1, 1\}$$

$$\text{DFT}[x_1 + x_2] = \{4, 0, 0, 0\}$$